

**DOUGLAS PLACE
LONG PLAT
DIVISION 2**

**BIORETENTION
DESIGN AND MAINTENANCE
CRITERIA**



05055
06036

**Homeowner's Association &
To be attached to the Purchase and Sale Agreement**

TABLE OF CONTENTS

1. Design Criteria 3

1.1. Bioretention Planting Specifications..... 4

1.1.1. Special Considerations..... 5

1.2. Bioretention Soil Specifications 6

1.3. Bioretention Mulch Specifications 7

2. Maintenance..... 8

2.1. Bioretention Maintenance..... 8

TABLES

Table 1-1: Douglas Place Long Plat Raingarden Dimensions..... 4

FIGURES

Raingarden/Bioretention Detail 9

Raingarden Plan Selection..... 10

Facility Maintenance checklists..... 14

EXHIBITS

MN-01 - DRAINAGE SYSTEM MAINTENANCE SCHEMATIC ~~28~~ 29

MN-02 – GRAVEL TRENCH DETAIL ~~29~~ 30

MN-03 – RAINGARDEN & SITE LAYOUT ~~30~~ 31

Introduction

Guidelines presented in this manual provide homeowners and landscape professionals with the information needed to design and build the required stormwater bioretention (raingardens) water quality facilities. Design criteria has been based on requirements per the Washington State Department of Ecology's 2005 *Stormwater Management Manual for Western Washington* (SWMM) and the Puget Sound Action Team's *Low Impact Development Technical Guidance Manual for Puget Sound* (LIDTGM). This manual also provides the maintenance and operational guidelines for the individual bioretention (raingarden) stormwater facilities.

A number of lots in Douglas Place Long Plat Div. 2 will treat, detain and infiltrate stormwater on site within raingardens, thereby reducing watershed impacts by retaining rainfall where it falls. Homeowners will provide water quality treatment and flow control by installing raingardens per specifications within this manual. Stormwater facilities (raingardens) have been sized for each lot based on lot area, infiltration rates, and an impervious area equal to 3,000 sf.

Joint stormwater facilities sized for the entire Douglas Place Long Plats, and South Douglas Long Plat include a wetpond, stormwater detention facility, control structure(s), and catch basins and conveyance piping. These facilities are described in the Stormwater Site Plan for Douglas Place Long Plat and Douglas Long Plat, Volumes I and II, dated May 2008.

An operation and maintenance document titled '*Douglas Place Long Plat & Douglas Long Plat Bioretention Design and Stormwater Facility Maintenance Criteria*', dated January 2008, addresses the joint stormwater facilities maintenance requirements.

A homeowner's association will own and maintain the joint stormwater facilities (South Douglas Long Plat recorded under Auditor File Number #2090102034 & Douglas Place Long Plat recorded under Auditor File Number #2131002304).

Individual raingardens will be owned and maintained by the property owners and/or the homeowner's association. These requirements are outlined on the Douglas Place Long Plat Div.2 plat, and recorded under Auditor File Number # 2131002302, at the Whatcom County Auditor's Office.

1. Design Criteria

As dictated by easements located on the final plat drawings, homeowners shall install bio-retention facilities based on the following design criteria. Bioretention areas (also referred to as rain gardens) will be integrated to detain and treat stormwater runoff from lawns and impervious surfaces. Rain gardens function as soil and plant-base facilities that remove pollutants through a variety of physical, biological, and chemical treatment processes. Raingardens will require watering until plantings are established. Figures C-1, C-2, C-3, and C-4 provide construction details and plant lists. Table 1-1 lists the raingarden areas and dimensions for each lot in the Douglas Place Long Plat Div. 2. Exhibit MN-1 provides a schematic of the drainage system that will require maintenance. Exhibit MN-2 is a copy of the Raingarden/Bioretention Detail (C-1) that shows a sectional view of a constructed raingarden with riser and over-flow elevations. Exhibit MN-3 is the Gravel Trench Detail of the sub-surface interception drain running south along the east side of the plat along lots 13, 12, 11, 10, 9, 8, 7, and 6. Exhibit MN-4 shows the lots required to have raingardens, dimensions and square footage of the raingardens. Construction drawing D-3 provides the riser elevations for the raingardens detail.

MN-5 an MN-6 shows preliminary lot numbering used in the earlier stormwater reports with areas not currently plated that require the use of raingardens. These raingarden areas may vary based on the final plat drawings. Refer to final design of plat for all raingarden sizing.

Table 1-1: Douglas Place Long Plat Raingarden Dimensions

Lot #	Area (acres)	Landscape (acres)	Infiltration (in/hr)	Bioretention (sf)	Gravel Depth (ft)	Total Depth (ft)	Depth to Groundwater (ft)
30	0.155	0.087	1.00	229	2	4	5
34	0.152	0.084	0.75	256	3	5	13
35	0.151	0.083	0.75	256	3	5	13
36	0.178	0.110	1.00	256	2.5	4.5	13

1.1. Bioretention Planting Specifications

The following text is referenced in Chapter 6 and Appendix 3 of the LIDTGM.

The primary design considerations for plant selection include:

- **Soil moisture conditions:** Plants should be tolerant of summer drought, ponding fluctuations, and saturated soil conditions for the lengths of time anticipated by the facility design.
- **Expected pollutant loadings:** Plants should tolerate typical pollutants and loadings from the surrounding land uses.
- **Above and below ground infrastructure in and near the facility:** Plant size and wind firmness should be considered within the context of the surrounding infrastructure. Rooting depths should be selected not to damage underground utilities if present.
- **Adjacent plant communities and potential invasive species control.**
- **Site distances and setbacks for roadway applications.**
- **Visual buffering:** Plants can be used to buffer structures from roads, enhance privacy among residences, and provide an aesthetic amenity for the site.
- **Aesthetics:** Visually pleasing plant designs add value to the property and encourage community and homeowner acceptance.

Native plant species, placed appropriately, tolerate local climate and biological stresses and usually require no nutrient or pesticide application in properly designed soil mixes. Natives can be used as the exclusive material in a rain garden or in combination with hardy cultivars that are not invasive and do not require chemical inputs.

To increase survival rates and ensure quality of plant material, the following guidelines are suggested:

- *All plant materials should have normal well-developed branches and vigorous root systems, and be free from physical defects, plant diseases, and insect pests.*
- *Plant size: Bioretention areas provide excellent soil conditions and should have well defined maintenance agreements. In this type of environment small plant material provides several advantages and is recommended. Small plant material requires less care in handling, less initial irrigation, experiences less transplant shock, is less expensive, adapts more quickly to a site, and transplants more successfully than larger material. Small trees and shrubs are also suggested.*

Bioretention cells and swales generally feature three planting zones characterized by soil moisture and periodic inundation.

- Zone 1:** *Area of periodic or frequent standing or flowing water. Zone 1 plants will also tolerate the seasonally dry periods of summer in the Pacific Northwest without extra watering and may also be applicable in Zone 2 or 3.*
- Zone 2:** *Periodically moist or saturated during larger storms. Plants listed under Zone 2 will also be applicable in Zone 3.*
- Zone 3:** *Dry soils, infrequently subject to inundation or saturation. This area can be used to transition or blend with the existing landscape.*

Appendix 1 contains the planting list for Zones 1-3.

1.1.1. Special Considerations

Drought tolerance – *Several plants included on the list do not tolerate dry conditions. For these plants, irrigation will be necessary during dry periods. In general, all plantings require watering during dry periods for the first two or three years after planting until established.*

Placement of large trees – *Consider height, spread, and extent of roots at maturity. Use caution in plant selection for areas with under-drain pipes or other structures. Lower limbs of plants placed close to a road or driveway may cause problems with visibility or safety.*

Appendix 2 contains the LIDTGM’s Street Tree List for more information on tree selection and placement suggestions.

Phytoremediation – *Appendix 3 includes a list of plants that have been studied for their ability to filter, absorb, and/or degrade specific contaminants. While most of these plants are not included in the following lists, varieties of some of the species known for phytoremediation are listed.*

1.2. Bioretention Soil Specifications

Either an on-site excavated soil mixed with compost (Bioretention Soil Mix 1) or a soil mixed off-site and delivered (Bioretention Soil Mix 2) shall be installed at each bioretention facility.

1. Bioretention Soil Mix 1

Bioretention Soil Mix 1 uses on-site excavated soil mixed with compost.
Design infiltration rate: 0.3 to 1.0 inch/hour (varies with properties of native soils).

Soil Mix:

- Approximately 65% approved on-site soil and 35% compost material thoroughly mixed.
- Excavated soil for mixing should be free of large woody debris or garbage (concrete or asphalt chunks, old pipe, etc).
- Collect and test representative samples of excavated soil for gradation.
- Using on-site excavated soil is not appropriate for on-site soils with high clay content. The excavated soil should be sandy loam, loamy sand or loam texture (USDA texture triangle). The excavated soil can be amended with appropriate aggregate (e.g. sand) to achieve the appropriate texture.
- Cover and store soil accordingly to prevent wetting or saturation.
- Test soil for fertility and micronutrients and, if necessary, amend mixture to create optimum conditions for plant establishment and early growth at rates recommended by an independent laboratory soil test.
- Organic content of the soil mixture should be 8% to 12%.

2. Bioretention Soil Mix 2

Bioretention Soil Mix 2 is mixed off-site and delivered ready for installation.
Design infiltration rate: 1 inch/hour.

Soil Mix:

- 65% to 70% gravelly sand and 30% to 35% compost (see specification below).
- Gravelly sand gradation per ASTM D 422.

Sieve size	Percent Passing
US No. 4	100
US No. 6	88-100
US No. 8	79-97
US No. 50	11-35
US No. 200	5-15
- Maximum clay content should be less than 5%.
- Soil mixture should be uniform, free of stones, stumps, roots or other similar objects larger than 2 inches.
- On-site soil mixing or placement not allowed if soil is saturated or subjected to water within 48 hours.
- Cover and store soil accordingly to prevent wetting or saturation.

- Test soil for fertility and micronutrients and, if necessary, amend mixture to create optimum conditions for plant establishment and early growth at rates recommended by an independent laboratory soil test.
- Organic content of the soil mixture should be 8% to 12%.

1.3. Bioretention Mulch Specifications

A 2 to 3 inch layer of mulch, with the following criteria shall be placed on bioretention facilities that are lacking a dense groundcover.

The following text is referenced in the LIDTGM Chapter 6.

The mulch layer can be easily removed or added to as part of standard and periodic landscape maintenance procedures. Properly selected mulch material reduces weed establishment, regulates soil temperatures and moisture, and adds organic matter to soil. Mulch when used, mulch should be:

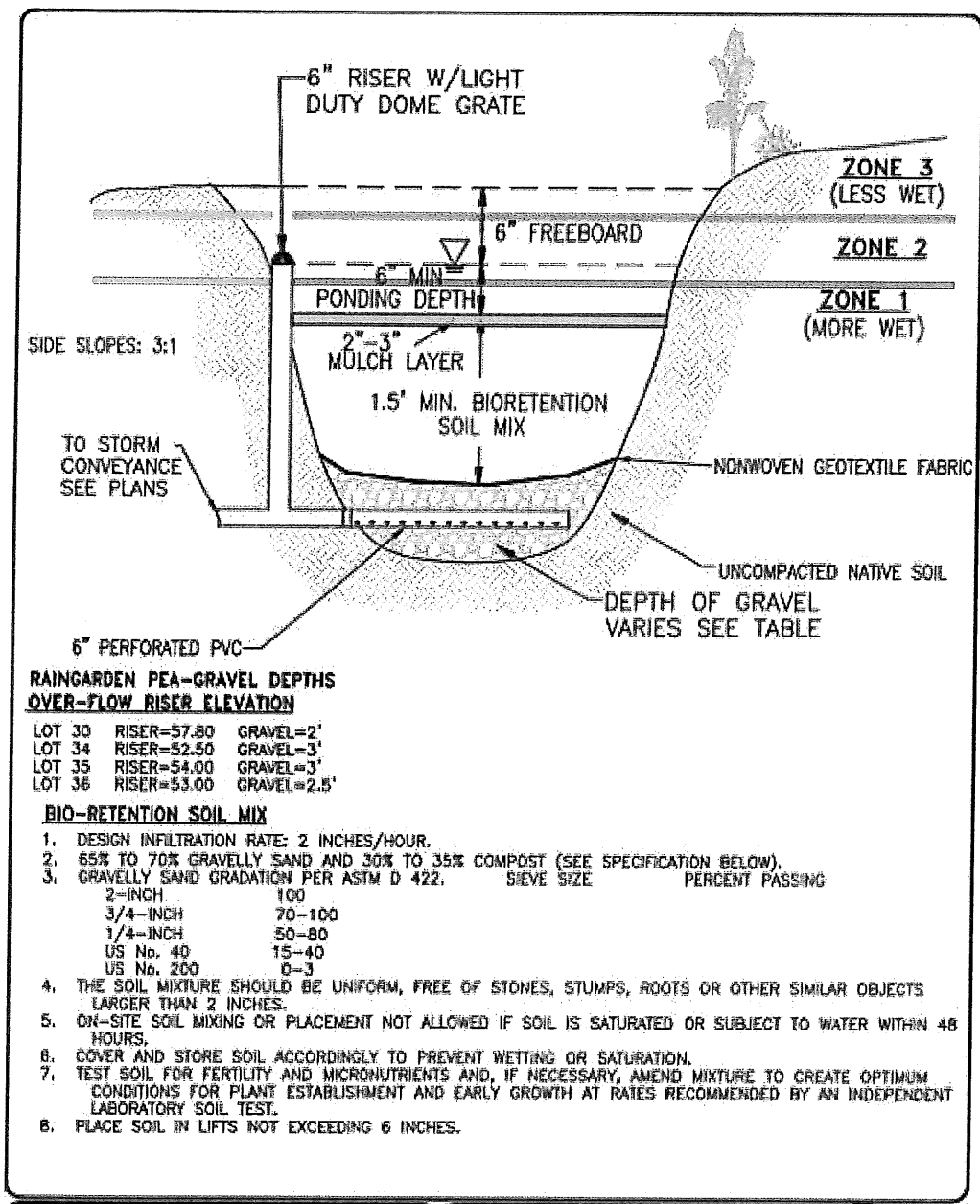
- *Compost in the bottom of the facilities (compost is less likely to float and is a better source for organic materials) and shredded or chipped hardwood or softwood in surrounding areas.*
- *Free of weed seeds, soil, roots and other material that is not bole (trunk of a tree) or branch wood and bark.*

2. Maintenance

2.1. Bioretention Maintenance

Bioretention areas require annual plant, soil, and mulch layer maintenance to ensure optimum infiltration, storage, and pollutant removal capabilities. In general, bioretention maintenance requirements are typical landscape care procedures and include:

- *Watering:* Plants should be selected to be drought tolerant and not require watering after establishment (2 to 3 years). Watering may be required during prolonged dry periods after plants are established.
- *Erosion control:* Inspect flow entrances, ponding area, and surface overflow areas periodically, and replace soil, plant material, and/or mulch layer in areas if erosion has occurred. Properly designed facilities with appropriate flow velocities should not have erosion problems except perhaps in extreme events. If erosion problems occur the following should be reassessed: (1) flow volumes from contributing areas and bioretention cell sizing; (2) flow velocities and gradients within the cell; and (3) flow dissipation and erosion protection strategies in the pretreatment area and flow entrance. If sediment is deposited in the bioretention area, immediately determine the source within the contributing area, stabilize, and remove excess surface deposits.
- *Plant material:* Depending on aesthetic requirements, occasional pruning and removing of dead plant material may be necessary. Replace all dead plants and if specific plants have a high mortality rate, assess the cause and replace with appropriate species. Periodic weeding is necessary until plants are established. The weeding schedule should become less frequent if the appropriate plant species and planting density have been used and, as a result, undesirable plants excluded.
- *Nutrient and pesticides:* The soil mix and plants are selected for optimum fertility, plant establishment, and growth. Nutrient and pesticide inputs should not be required and may degrade the pollutant processing capability of the bioretention area, as well as contribute pollutant loads to receiving waters. By design, bioretention facilities are located in areas where phosphorous and nitrogen levels are often elevated and these should not be limiting nutrients. If in question, have soil analyzed for fertility.
- *Mulch:* Replace mulch annually in bioretention facilities where heavy metal deposition is likely (e.g., contributing areas that include parking lots and roads). In residential lots or other areas where metal deposition is not a concern, replace or add mulch as needed to maintain a 2 to 3 inch depth at least once every two years.
- *Soil:* Soil mixes for bioretention facilities are designed to maintain long-term fertility and pollutant processing capability. Estimates from metal attenuation research suggest that metal accumulation should not present an environmental concern for at least 20 years in bioretention systems. Replacing mulch in bioretention facilities where heavy metal deposition is likely provides an additional level of protection for prolonged performance. If in question, have soil analyzed for fertility and pollutant levels.



NOTE: THE CONSTRUCTED RAINGARDEN/BIORETENTION SHALL BE INSPECTED BY THE ENGINEER OF RECORD PRIOR TO OCCUPANCY

FIGURE C-1 RAINGARDEN/BIORETENTION DETAIL
DOUGLAS PLACE DIVISION 2 MAINTENANCE PLAN

NOTES FOR RAINGARDEN CONSTRUCTION:

- The site preparation for each Raingarden is uniformly the same and is as follows:
- 1. The site will need to be excavated to a depth of 34" to set the grade for the installation of the 6" perforated pipe with 8" of pea gravel that is tied to the stormwater drainage system. (See Civil Plans).
 - 2. In addition, a 1.5' layer of Bioretention Soil Mix is spread on the surface area. To develop the zones II and III - the average additional depth per zone is 3-4 " approximately. Lastly, organic compost material will be spread on the area approximately 3-4 inches.
 - 3. Plant Schedule needs to be implemented between November 1st and March 31st.

RAINGARDEN: Zone I PLANT SCHEDULE RAINGARDEN TOTAL AREA = 25' x 10.8' = 270 s.f. ZONE I: TOTAL AREA = 80 S.F. OR 30% OF AREA					
Zone I: Most wet; inundated	Species Options	Size	Density	Number	Cost
TREES: < 2%					
Hooker Willow (FACW)	Salix hookeriana	36"	4 'center		
Bog Birch (OBL)	Betula glandulosa	36"	4' center		
Oregon Ash (FACW)	Fraxinus latifolia	36"	4' center		
Pacific Crabapple (FACW)	Malus fusca	36"	4 'center	1	
SHRUBS: 25-35%					
Red Osier Dogwood (FACW)	Cornus stolonifera	24"	3' center *clusters	7	
Pacific Bayberry (FACW)	Myrica gale	24"	3' center	2	
Pacific Ninebark (FACW)	Physocarpus capitatus	24"	3' center		
HERBACEOUS: 45%-75%					
Slough Sedge (OBL)	Carex obnupta	Plugs	18"center	15	
Largeleaf Avens (FACW)	Geum macrophyllum	Plugs	18"center	10	
Skunk Cabbage (OBL)	Lysichiton americanum	Plugs	18"center		
Total Plants				35	
Soil Amendment:				See Civil Plans	
Mulching Installed:				1 YARD @ 4" FOR 80 S.F.	

FIGURE C-2 RAINGARDEN ZONE I - PLANT SCHEDULE

RAINGARDEN: Zone II PLANT SCHEDULE RAINGARDEN TOTAL AREA = 25' x 10.8' = 270 s.f. ZONE II TOTAL AREA = 165 S.F. OR 60% OF TOTAL AREA					
Zone II: slope of raingarden: frequent, variable inundation	Species Options:	Size	Density	Number	Cost
TREES: < 11-15%					
Quaking Aspen (FAC)	Populus tremuloides	36"	4' center	5	
Western Red Cedar (FAC)	Thuja plicata	36"	4' center		
Black Hawthorne (FAC)	Crataegus douglasii	36"	4' center		
SHRUBS: 60%-70%					
Nootka Rose (FAC)	Rosa nutkana	24"	3' center *clusters	10	
Salmon berry (FAC+)	Rubus spectabilis	24"	3' centers		
Twinberry (FAC+)	Lonicera involucrata	24"	3' center	10	
Swamp Gooseberry (FAC+)	Ribes lacustre	24"	3' center	7	
HERBACEOUS: 25%-30%					
Bunchberry (FAC)	Cornus Canadensis	Bulb	18" center	12	
White Clover (FAC)	Trifolium repens	Seed	Dispersed	1 lb	
Total Plants				44	
Soil Amendment				See Civil Plans	
Mulching Installed				2 YARDS TO COVER AREA OF 165 S.F. WITH 4"	

FIGURE C-3 RAINGARDEN ZONE II - PLANT SCHEDULE

RAINGARDEN - Zone III					
PLANT SCHEDULE					
RAINGARDEN TOTAL AREA = 25' x 10.8' = 270 s.f.					
ZONE III TOTAL AREA = +/- 25 S.F. OR 10% OF TOTAL AREA					
Zone III: Slope: namely drought tolerant	Species Options	Size	Density	Number	Cost
TREES: 1%-10%					
Scouler Willow (FAC)	Salix scouleriana	36"	4' center		
Beaked Hazelnut (FACU)	Corylus cornuta	36"	4' center	1	
Cascara (FAC-)	Rhamnus purshiana	36"	4' center		
SHRUBS: 10%-90%					
Vine Maple (FAC-)	Acer circinatum	24"	3' center	4	
Common Snowberry (FACU)	Symphoricarpos alba	24"	3' center		
Oregon Grape (FACU)	Mahonia nervosa	24"	3' center	2	
Red Flowering Currant (UPL)	Ribes sanguineum	24"	3' center	4	
Indian Plum (FACU)	Oemleria cerasiformis	24"	3' center		
Red Elderberry (FACU)	Sambucus racemosa	24"	3' center		
HERBACEOUS					
Colonial Bentgrass (FAC)	Agrostis capillaris	Seed	Dispersed	Note: 1 lb of <i>Native seed grass only</i> to cover slope. Will need to cover with weed-free certified straw for 1 st season.	
Redtop	Agrostis gigantea	Seed	Dispersed		
Red Clover (FACU)	Trifolium pratense	Seed	Dispersed		
Total Plants				11	
Soil Amendment				See Civil Plans	
Mulching Installed				¼ YARD FOR AREA: 25 S.F. @ 4"	

FIGURE C-4 RAINGARDEN ZONE III - PLANT SCHEDULE

3. Stormwater Management Facilities

To help understand stormwater facility maintenance requirements, it is useful to have a general knowledge of how they function. Some maintenance needs are common to all types of facilities, while others depend on the specific facility.

The three major components of stormwater management include stormwater collection/conveyance, stormwater quantity control (detention/retention) and stormwater quality control (treatment). This section describes general stormwater management theories and goals. Specific stormwater facility descriptions and maintenance requirements are provided in Section 3.0.

3.1. Collection and Conveyance Systems

Collection and conveyance systems intercept and transport stormwater and typically consist of inlets that collect water and pipes and/or open channels (ditches). Stormwater conveyance systems are designed to provide capacity for a specific maximum flow rate. Typical failures include reduced capacity due to clogged surface grates and pipes. Plugging commonly occurs due to sediment and large debris washed from adjacent surfaces. Reduced conveyance system capacity results in localized flooding and possible property damage.

3.2. Stormwater Quantity Control (Detention/Retention)

The intent of stormwater quantity control facilities is to slow down stormwater flow discharged to the environment from developed sites. Impervious surfaces, such as roads, roofs, and lawns, quicken the rate of stormwater runoff into natural streams which can create flooding. Stormwater quantity control facilities mitigate the increased runoff by providing temporary storage and controlling the release rate from the site to prevent flooding and erosion. Detention and retention facilities may be designed as ponds or raingardens.

Detention facilities function by providing temporary storage of stormwater runoff to be released at a controlled rate. The intent of the detention facility is to match the pre-developed runoff rates for several specific storm events in the developed condition.

Retention facilities are typically located in areas where water soaks easily into the ground. Retention facilities provided temporary storage while allowing the water to soak into the ground, mimicking natural conditions. There is typically no release of stormwater to other pipes or water bodies.

3.3. Stormwater Quality Control (Treatment)

There are a several Best Management Practices (BMPs) utilized for stormwater quality control. These systems provide stormwater treatment through a combination of filtration, sediment settling, plant nutrient uptake, and physical separation. The most common treatment systems include raingardens, biofiltration swales, filter strips, wetponds, and sandfilters. The intent of all stormwater treatment facilities is to remove oils, chemicals, metals, and sediment from stormwater runoff prior to being discharged from the property.

Stormwater treatment facilities have a limited pollutant removal capability and are not intended to replace proper site management. The most effective technique for reducing pollutant discharge from the site is to provide good housekeeping through source control Best Management Practices (BMPs) as provided in the Stormwater Pollution Prevention Manual located at <http://dnr.metrokc.gov/wlr/stormwater>

4. Facility Descriptions and Maintenance Checklists

Each of the following subsections includes a facility description, illustrated exhibit, and maintenance checklist for 8 stormwater facilities/components utilized in the Douglas Place Long Plat. These include:

- Detention Ponds
- Control Structure/ Flow Restrictor
- Catch Basins
- Debris Barriers (e.g. Trash Racks)
- Wetponds
- Fencing/Shrubbery Screen/Other Landscaping
- Gates
- Grounds (Landscaping)

Private stormwater facility owners should print the checklists that apply to their facilities and follow the inspection recommendations and frequencies. The checklists should be used for the maintenance report submitted to the City as required by maintenance agreement or for storm utility credit.

4.1. Detention Ponds

Stormwater detention ponds are open basins built by excavating below existing ground or by constructing above-ground berms (embankments). The detention pond temporarily stores stormwater runoff during rain events and slowly releases it through an outlet (control structure). Detention ponds are typically designed to completely drain within 24 hours after the completion of a storm event. Components that are typically associated with a detention pond include the following: control structure/flow restrictor, debris barrier (e.g. trash rack), energy dissipaters, access road, and fence.

Detention Ponds Checklist

Frequency	Drainage System Feature	Date				Problem	Conditions to Check For	Conditions That Should Exist
		✓	✓	✓	✓			
M,S	General					Trash & Debris	Any trash and debris which exceed 5 cubic feet per 1,000 square feet (this is about equal to the amount of trash it would take to fill up one standard size garbage can). In general, there should be no visual evidence of dumping. If less than threshold all trash and debris will be removed as part of next scheduled maintenance.	Trash and debris cleared from site.
A	General					Poisonous Vegetation and noxious weeds	Any poisonous or nuisance vegetation which may constitute a hazard to maintenance personnel or the public. Any evidence of noxious weeds as defined by State or local regulations. (Apply requirements of adopted IPM policies for the use of herbicides).	No danger of poisonous vegetation where maintenance personnel or the public might normally be. (Coordinate with local health department) Complete eradication of noxious weeds may not be possible. Compliance with State or local eradication policies required
M,S	General					Contaminants and Pollution	Any evidence of oil, gasoline, contaminants or other pollutants	No contaminants or pollutants present. (Coordinate removal/cleanup with local water quality response agency).
M	General					Rodent Holes	Any evidence of rodent holes if facility is acting as a dam or berm, or any evidence of water piping through dam or berm via rodent holes.	Rodents destroyed and dam or berm repaired. (Coordinate with local health department; coordinate with Ecology Dam Safety Office if pond exceeds 10 acre-feet.)
M	General					Beaver Dams	Dam results in change or function of the facility.	Facility is returned to design function. (Coordinate trapping of beavers and removal of dams with appropriate permitting agencies)
A	General					Insects	When insects such as wasps and hornets interfere with maintenance activities.	Insects destroyed or removed from site. Apply insecticides in compliance with adopted IPM policies.
A	General					Tree Growth and Hazard Trees	Tree growth does not allow maintenance access or interferes with maintenance activity (i.e., slope mowing, silt removal, vactoring, or equipment movements). If trees are not interfering with access or maintenance, do not remove	Trees do not hinder maintenance activities. Harvested trees should be recycled into mulch or other beneficial uses (e.g., alders for firewood).
A	General						If dead, diseased, or dying	Remove hazard Trees

Frequency	Drainage System Feature	Date				Problem	Conditions to Check For	Conditions That Should Exist
		✓	✓	✓	✓			
							trees are identified (Use a certified Arborist to determine health of tree or removal requirements)	
M	Side Slopes of Pond					Erosion	Eroded damage over 2 inches deep where cause of damage is still present or where there is potential for continued erosion.	Slopes should be stabilized using appropriate erosion control measure(s); e.g., rock reinforcement, planting of grass, compaction.
M,S	Side Slopes of Pond						Any erosion observed on a compacted berm embankment.	If erosion is occurring on compacted berms a licensed civil engineer should be consulted to resolve source of erosion.
M	Storage Area					Sediment	Accumulated sediment that exceeds 10% (typically 6" to 12") of the designed pond depth unless otherwise specified or affects inletting or outletting condition of the facility.	Sediment cleaned out to designed pond shape and depth; pond reseeded if necessary to control erosion.
M	Storage Area					Liner (If Applicable)	Liner is visible and has more than three 1/4-inch holes in it.	Liner repaired or replaced. Liner is fully covered.
A	Pond Berms (Dikes)					Settlements	Any part of berm which has settled 4 inches lower than the design elevation. If settlement is apparent, measure berm to determine amount of settlement. Settling can be an indication of more severe problems with the berm or outlet works. A licensed civil engineer should be consulted to determine the source of the settlement.	Dike is built back to the design elevation.
A	Pond Berms (Dikes)					Piping	Discernable water flow through pond berm. Ongoing erosion with potential for erosion to continue. (Recommend a Geotechnical engineer be called in to inspect and evaluate condition and recommend repair of condition.	Piping eliminated. Erosion potential resolved.

Frequency	Drainage System Feature	Date				Problem	Conditions to Check For	Conditions That Should Exist
		✓	✓	✓	✓			
A	Emergency Overflow/ Spillway					Tree Growth	Tree growth on emergency spillways creates blockage problems and may cause failure of the berm due to uncontrolled overtopping.	Trees should be removed. If root system is small (base less than 4 inches) the root system may be left in place. Otherwise the roots should be removed and the berm restored. A licensed civil engineer should be consulted for proper berm/spillway restoration.
A	Emergency Overflow/ Spillway					Emergency Overflow/ Spillway	Only one layer of rock exists above native soil in area five square feet or larger, or any exposure of native soil at the top of out flow path of spillway. (Rip-rap on inside slopes need not be replaced.)	Rocks and pad depth are restored to design standards.

If you are unsure whether a problem exists, please contact a Professional Engineer.

Comments:

Key:

- (M) Monthly from November through April.
- (A) Once in late summer (preferable September)
- (S) After any major storm (use 1-inch in 24 hours as a guideline).

4.2. Control Structure/Flow Restrictor

Control structures/flow restrictors are located on the outlet pipe of a detention system. The control structure is typically a Type 2 concrete catch basin (see Section 3.5 for catch basin description) with a riser (vertical pipe). The control structure reduces the discharge rate of stormwater from a detention facility. The flow is regulated by a combination of orifices (holes with specifically sized diameters) and weirs (plates with rectangular or vee shaped notch). Lack of maintenance of the control structure can result in the plugging of an orifice. This can result in flooding of the stormwater system and/or an increase in the rate of discharge from the site potentially damaging downstream property.

Control Structure/Flow Restrictor Checklist

Frequency	Drainage System Feature	Date				Problem	Conditions to Check For	Conditions That Should Exist
		✓	✓	✓	✓			
M	General					Trash and Debris (Includes Sediment)	Material exceeds 25% of sump depth or 1 foot below orifice plate.	Control structure orifice is not blocked. All trash and debris removed.
A	General					Structural Damage	Structure is not securely attached to manhole wall.	Structure securely attached to wall and outlet pipe.
A	General					Structural Damage	Structure is not in upright position (allow up to 10% from plumb).	Structure in correct position.
A	General					Structural Damage	Connections to outlet pipe are not watertight and show signs of rust.	Connections to outlet pipe are water tight; structure repaired or replaced and works as designed.
A	General					Structural Damage	Any holes--other than designed holes--in the structure.	Structure has no holes other than designed holes.
A	Cleanout Gate					Damaged or Missing	Cleanout gate is not watertight or is missing.	Gate is watertight and works as designed.
A	Cleanout Gate					Damaged or Missing	Gate cannot be moved up and down by one maintenance person.	Gate moves up and down easily and is watertight.
A	Cleanout Gate					Damaged or Missing	Chain/rod leading to gate is missing or damaged.	Chain is in place and works as designed.
A	Cleanout Gate					Damaged or Missing	Gate is rusted over 50% of its surface area.	Gate is repaired or replaced to meet design standards.
A	Orifice Plate					Damaged or Missing	Control device is not working properly due to missing, out of place, or bent orifice plate.	Plate is in place and works as designed.
M,S	Orifice Plate					Obstructions	Any trash, debris, sediment, or vegetation blocking the plate.	Plate is free of all obstructions and works as designed.
	Overflow Pipe					Obstructions	Any trash or debris blocking (or having the potential of blocking) the overflow pipe.	Pipe is free of all obstructions and works as designed.

Frequency	Drainage System Feature	Date				Problem	Conditions to Check For	Conditions That Should Exist
		✓	✓	✓	✓			
A	Manhole					Cover Not in Place	Cover is missing or only partially in place. Any open manhole requires maintenance.	Manhole is closed.
A	Manhole					Locking Mechanism Not Working	Mechanism cannot be opened by one maintenance person with proper tools. Bolts into frame have less than 1/2 inch of thread (may not apply to self-locking lids).	Mechanism opens with proper tools.
A	Manhole					Cover Difficult to Remove	One maintenance person cannot remove lid after applying normal lifting pressure. Intent is to keep cover from sealing off access to maintenance.	Cover can be removed and reinstalled by one maintenance person.
A	Manhole					Ladder Rungs Unsafe	Ladder is unsafe due to missing rungs, misalignment, not securely attached to structure wall, rust, or cracks.	Ladder meets design standards. Allows maintenance person safe access.

If you are unsure whether a problem exists, please contact a Professional Engineer.

Comments:

Key:

- (M) Monthly from November through April.
- (A) Once in late summer (preferable September)
- (S) After any major storm (use 1-inch in 24 hours as a guideline).

4.3. Catch Basins

Catch basins are underground concrete structures typically provided with a slotted grate to collect stormwater runoff and route it through underground pipes. Catch basins can also be used as a junction in a pipe system and may have a solid lid. There are two catch basin types.

A Type 1 catch basin is a rectangular box with approximate dimensions of 3’x2’x5’. Type 1 catch basins are utilized when the connected conveyance pipes are less than 18 inches in diameter and the depth from the gate to the bottom of the pipe is less than 5 feet.

Type 2 catch basins, also commonly referred to as storm manholes, are round concrete structures ranging in diameter of 4 feet to 8 feet. Type 2 catch basins are used when the connecting conveyance pipe is 18 inches or greater or the depth from grate to pipe bottom exceeds 5 feet. Type 2 catch basins typically have manhole steps mounted on the side of the structure to allow for access.

Both catch basin types typically provide a storage volume (sump) below the outlet pipe to allow sediments and debris to settle out of the stormwater runoff. Some catch basins are also provided with a spill control device (inverted elbow on outlet pipe) intended to contain large quantities of grease or oils.

The most common cleaning method for catch basins is to utilize a truck with a tank and vacuum hose (vactor truck) to remove sediment and debris from the sump. Catch basins may be an enclosed space where harmful chemicals and vapors can accumulate. Therefore, if the inspection and maintenance requires entering a catch basin, it should be conducted by an individual with training and certification in working in hazardous confined spaces.

Catch Basins Checklist

Frequency	Drainage System Feature	Date				Problem	Conditions to Check For	Conditions That Should Exist
		✓	✓	✓	✓			
A	General					“Dump no pollutants” “ Stencil or stamp not visible	Stencil or stamp should be visible and easily read	Warning signs (e.g., "Dump No Waste-Drains to Stream") shall be painted or embossed on or adjacent to all storm drain inlets.
M,S	General					Trash & Debris	Trash or debris which is located immediately in front of the catch basin opening or is blocking inletting capacity of the basin by more than 10%.	No trash or debris located immediately in front of catch basin or on grate opening.
M	General					Trash & Debris	Trash or debris (in the basin) that exceeds 60 percent of the sump depth as measured from the bottom of basin to invert of the lowest pipe into or out of the basin, but in no case less than a minimum of six inches clearance from the debris surface to the invert of the lowest pipe.	No trash or debris in the catch basin.

Frequency	Drainage System Feature	Date				Problem	Conditions to Check For	Conditions That Should Exist
		✓	✓	✓	✓			
M	General					Trash & Debris	Trash or debris in any inlet or outlet pipe blocking more than 1/3 of its height.	Inlet and outlet pipes free of trash or debris.
M	General					Trash & Debris	Dead animals or vegetation that could generate odors that could cause complaints or dangerous gases (e.g., methane).	No dead animals or vegetation present within the catch basin.
M	General					Sediment	Sediment (in the basin) that exceeds 60 percent of the sump depth as measured from the bottom of basin to invert of the lowest pipe into or out of the basin, but in no case less than a minimum of 6 inches clearance from the sediment surface to the invert of the lowest pipe.	No sediment in the catch basin
A	General					Structure Damage to Frame and/or Top Slab	Top slab has holes larger than 2 square inches or cracks wider than 1/4 inch (Intent is to make sure no material is running into basin).	Top slab is free of holes and cracks.
A	General					Structure Damage to Frame and/or Top Slab	Frame not sitting flush on top slab, i.e., separation of more than 3/4 inch of the frame from the top slab. Frame not securely attached.	Frame is sitting flush on the riser rings or top slab and firmly attached.
A	General					Fractures or Cracks in Basin Walls/Bottom	Maintenance person judges that structure is unsound.	Basin replaced or repaired to design standards.
A	General					Fractures or Cracks in Basin Walls/Bottom	Grout fillet has separated or cracked wider than 1/2 inch and longer than 1 foot at the joint of any inlet/outlet pipe or any evidence of soil particles entering catch basin through cracks.	Pipe is re-grouted and secure at basin wall.
A	General					Settlement / Misalignment	If failure of basin has created a safety, function, or design problem.	Basin replaced or repaired to design standards.
M	General					Vegetation	Vegetation growing across and blocking more than 10% of the basin opening.	No vegetation blocking opening to basin.
M	General					Vegetation	Vegetation growing in inlet/outlet pipe joints that is more than six inches tall and less than six inches apart.	No vegetation or root growth present.
M	General					Contamination and Pollution	Any evidence of oil, gasoline, contaminants, or other pollutants (Coordinate removal/cleanup with local water quality response agency).	No contaminants or pollutants present.

Frequency	Drainage System Feature	Date				Problem	Conditions to Check For	Conditions That Should Exist
		✓	✓	✓	✓			
A	Catch Basin Cover					Cover Not in Place	Cover is missing or only partially in place.	Any open catch basin requires maintenance. Catch basin cover is closed
A	Catch Basin Cover					Locking Mechanism Not Working	Mechanism cannot be opened by one maintenance person with proper tools. Bolts into frame have less than 1/2 inch of thread.	Mechanism opens with proper tools.
A	Catch Basin Cover					Cover Difficult to Remove	One maintenance person cannot remove lid after applying normal lifting pressure. (Intent is to keep cover from sealing off access to maintenance.)	Cover can be removed by one maintenance person.
A	Ladder					Ladder Rungs Unsafe	Ladder is unsafe due to missing rungs, not securely attached to basin wall, misalignment, rust, cracks, or sharp edges.	Ladder meets design standards and allows maintenance person safe access.
	Grates					Grate opening Unsafe	Grate with opening wider than 7/8 inch.	Grate opening meets design standards.
M,S	Grates					Trash and Debris	Trash and debris that is blocking more than 20% of grate surface inletting capacity.	Grate free of trash and debris.
A	Grates					Damaged or Missing.	Grate missing or broken member(s) of the grate.	Grate is in place and meets design standards.

If you are unsure whether a problem exists, please contact a Professional Engineer.

Comments:

Key:

- (M) Monthly from November through April.
- (A) Once in late summer (preferable September)
- (S) After any major storm (use 1-inch in 24 hours as a guideline).

4.4. Wetponds

A wetpond is an open basin that retains a permanent pool of water (wetpool) year round or only during the wet season. The volume of the wetpool allows sediment and other pollutants to settle out of the runoff. Wetland vegetation is typically planted within the wetpond to provide additional treatment through nutrient (i.e. nitrogen) removal. Detention quantity control can be provided with additional temporary storage volume above the permanent pool elevation.

Wetponds Checklist

Frequency	Drainage System Feature	Date				Problem	Conditions to Check For	Conditions That Should Exist
		✓	✓	✓	✓			
M	General					Water level	First cell is empty, doesn't hold water.	Line the first cell to maintain at least 4 feet of water. Although the second cell may drain, the first cell must remain full to control turbulence of the incoming flow and reduce sediment re-suspension.
M						Trash and Debris	Accumulation that exceeds 1 CF per 1000-SF of pond area.	Trash and debris removed from pond
M						Inlet/Outlet Pipe	Inlet/Outlet pipe clogged with sediment and/or debris material	No clogging or blockage in the inlet and outlet piping.
M						Sediment Accumulation in Pond Bottom	Sediment accumulations in pond bottom that exceeds the depth of sediment zone plus 6-inches, usually in the first cell.	Sediment removed from pond bottom.
M						Oil Sheen on Water	Prevalent and visible oil sheen.	Oil removed from water using oil- absorbent pads or vactor truck. Source of oil located and corrected. If chronic low levels of oil persist, plant wetland plants such as Juncus effusus (soft rush) which can uptake small concentrations of oil.
A						Erosion	Erosion of the pond's side slopes and/or scouring of the pond bottom, which exceeds 6-inches, or where continued erosion is prevalent.	Slopes stabilized using proper erosion control measures and repair methods.

Frequency	Drainage System Feature	Date				Problem	Conditions to Check For	Conditions That Should Exist
		✓	✓	✓	✓			
A						Settlement of Pond Dike/Berm	Any part of these components which has settled 4-inches or lower than the design elevation, or inspector determines dike/berm is unsound.	Dike/berm is repaired to specifications
A						Internal Berm	Berm dividing cells should be level.	Berm surface is leveled so that water flows evenly over entire length of berm.
A						Overflow Spillway	Rock is missing and soil is exposed at top of spillway or outside slope.	Rocks replaced to specifications.

If you are unsure whether a problem exists, please contact a Professional Engineer.

Comments:

Key:

- (M) Monthly from November through April.
- (A) Once in late summer (preferable September)
- (S) After any major storm (use 1-inch in 24 hours as a guideline).

4.5. Fencing/Shrubbery Screen/Other Landscaping

Fencing and shrubbery screen are provided around open stormwater management facilities to limit unauthorized access for safety purposes and to minimize the visual impact of the facility.

Fencing/Shrubbery Screen/Other Landscaping Checklist

Frequency	Drainage System Feature	Date				Problem	Conditions to Check For	Conditions That Should Exist
		✓	✓	✓	✓			
M	General					Missing or broken parts/dead shrubbery	Any defect in the fence or screen that permits easy entry to a facility.	Fence is mended or shrubs replaced to form a solid barrier to entry.
M,S	General					Erosion	Erosion has resulted in an opening under a fence that allows entry by people or pets.	Replace soil under fence so that no opening exceeds 4 inches in height.
M	General					Unruly vegetation	Shrubbery is growing out of control or is infested with weeds.	Shrubbery is trimmed and weeded to provide appealing aesthetics. Do not use chemicals to control weeds.
A	Fences					Damaged parts	Posts out of plumb more than 6 inches.	Posts plumb to within 1-1/2 inches of plumb.
A	Fences					Damaged parts	Top rails bent more than 6 inches.	Top rail free of bends greater than 1 inch.
A	Fences					Damaged parts	Any part of fence (including posts, top rails, and fabric) more than 1 foot out of design alignment.	Fence is aligned and meets design standards.
A	Fences					Damaged parts	Missing or loose tension wire.	Tension wire in place and holding fabric.
A	Fences					Damaged parts	Missing or loose barbed wire that is sagging more than 2-1/2 inches between posts.	Barbed wire in place with less than 3/4-inch sag between posts.
A	Fences					Damaged parts	Extension arm missing, broken, or bent out of shape more than 1-1/2 inches.	Extension arm in place with no bends larger than 3/4 inch.
A	Fences					Deteriorated paint or protective coating	Part or parts that have a rusting or scaling condition that has affected structural adequacy.	Structurally adequate posts or parts with a uniform protective coating.
M	Fences					Openings in fabric	Openings in fabric are such that an 8-inch diameter ball could fit through.	No openings in fabric.

If you are unsure whether a problem exists, please contact a Professional Engineer.

Key:

- (M) Monthly from November through April.
- (A) Once in late summer (preferable September)
- (S) After any major storm (use 1-inch in 24 hours as a guideline).

4.6. Gates

Gates typically consist of a chain link gate for fenced stormwater facilities to provide safety and allow vehicle and/or personnel access to the facility.

Gates Checklist

Frequency	Drainage System Feature	Date				Problem	Conditions to Check For	Conditions That Should Exist
		✓	✓	✓	✓			
M	General					Damaged or missing components	Gate is broken, jammed, or missing.	Pond has a functioning gate to allow entry of people and maintenance equipment such as mowers and backhoe. If a lock is used, make sure the County field staff has a key.
M	General					Damaged or missing components	Broken or missing hinges such that gate cannot be easily opened and closed by a maintenance person.	Hinges intact and lubed. Gate is working freely.
A	General					Damaged or missing components	Gate is out of plumb more than 6 inches and more than 1 foot out of design alignment.	Gate is aligned and vertical.
A	General					Damaged or missing components	Missing stretcher bands, and ties.	Stretcher bar, bands, and ties in place.

If you are unsure whether a problem exists, please contact a Professional Engineer.

Comments:

Key:

- (M) Monthly from November through April.
- (A) Once in late summer (preferable September)
- (S) After any major storm (use 1-inch in 24 hours as a guideline).

4.7. Grounds (Landscaping)

Landscaping is an essential component of stormwater management. Bare soil areas generate higher levels of stormwater runoff and sedimentation in stormwater facilities. The following check list gives some general guidance for landscape management.

Grounds (Landscaping) Checklist

Frequency	Drainage System Feature	Date				Problem	Conditions to Check For	Conditions That Should Exist
		✓	✓	✓	✓			
M	General					Weeds (nonpoisonous)	Weeds growing in more than 20% of the landscaped area (trees and shrubs only).	Weeds present in less than 5% of the landscaped area.
M	General					Insect hazard	Any presence of poison ivy or other poisonous vegetation or insect nests.	No poisonous vegetation or insect nests present in landscaped area.
M,S	General					Trash or litter	See Ponds Checklist.	See Ponds Checklist.
M,S	General					Erosion of Ground Surface	Noticeable rills are seen in landscaped areas.	Causes of erosion are identified and steps taken to slow down/spread out the water. Eroded areas are filled, contoured, and seeded.
A	Trees and shrubs					Damage	Limbs or parts of trees or shrubs that are split or broken which affect more than 25% of the total foliage of the tree or shrub.	Trim trees/shrubs to restore shape. Replace trees/shrubs with severe damage.
M	Trees and shrubs					Damage	Trees or shrubs that have been blown down or knocked over.	Replant tree, inspecting for injury to stem or roots. Replace if severely damaged.
A	Trees and shrubs					Damage	Trees or shrubs which are not adequately supported or are leaning over, causing exposure of the roots.	Place stakes and rubber-coated ties around young trees/shrubs for support.

If you are unsure whether a problem exists, please contact a Professional Engineer.

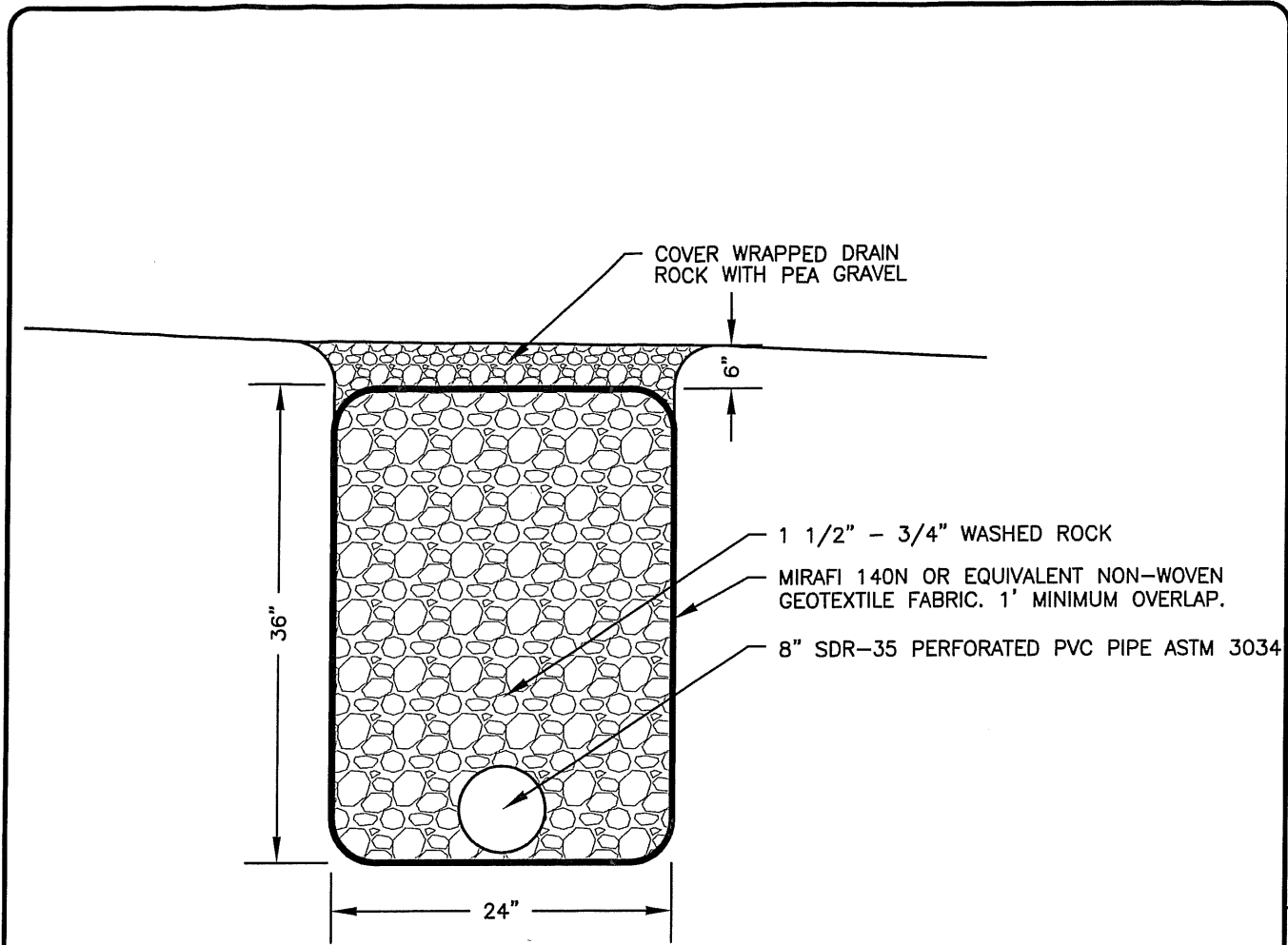
Comments:

- Key:
- (M) Monthly from November through April.
 - (A) Once in late summer (preferable September)
 - (S) After any major storm (use 1-inch in 24 hours as a guideline).

5. Developing a Maintenance Program

A stormwater maintenance program is essential to ensure that the facilities continue to function as designed to prevent possible flooding and property damage. The maintenance program consists of inspections and repairs as detailed in the maintenance checklists provided in Section 3.0.

Stormwater management facilities are not intended to replace good housekeeping procedures. Good housekeeping includes educating facility users of proper storage and disposal of chemicals and potential pollutants, procedures for spill cleanup, proper use of fertilizers and other lawn care products, and maintenance of equipment to prevent release of pollutants to the stormwater system. Guidelines for establishing good housekeeping procedures (Source Control BMPs) and developing a training program to educate facility users can be found in the King County Stormwater Pollution Prevention Manual located at: <http://dnr.metrokc.gov/wlr/dss/sppm.htm>



Page ³⁰~~36~~ ³¹ of 37

L:\Jepson Projects\2012\12082\Eng\Drawings\Douglas-Place_Div_2-Drainage\140130-12082AB.dwg
- MN-03 - sgoodrich - Jan, 30 2014

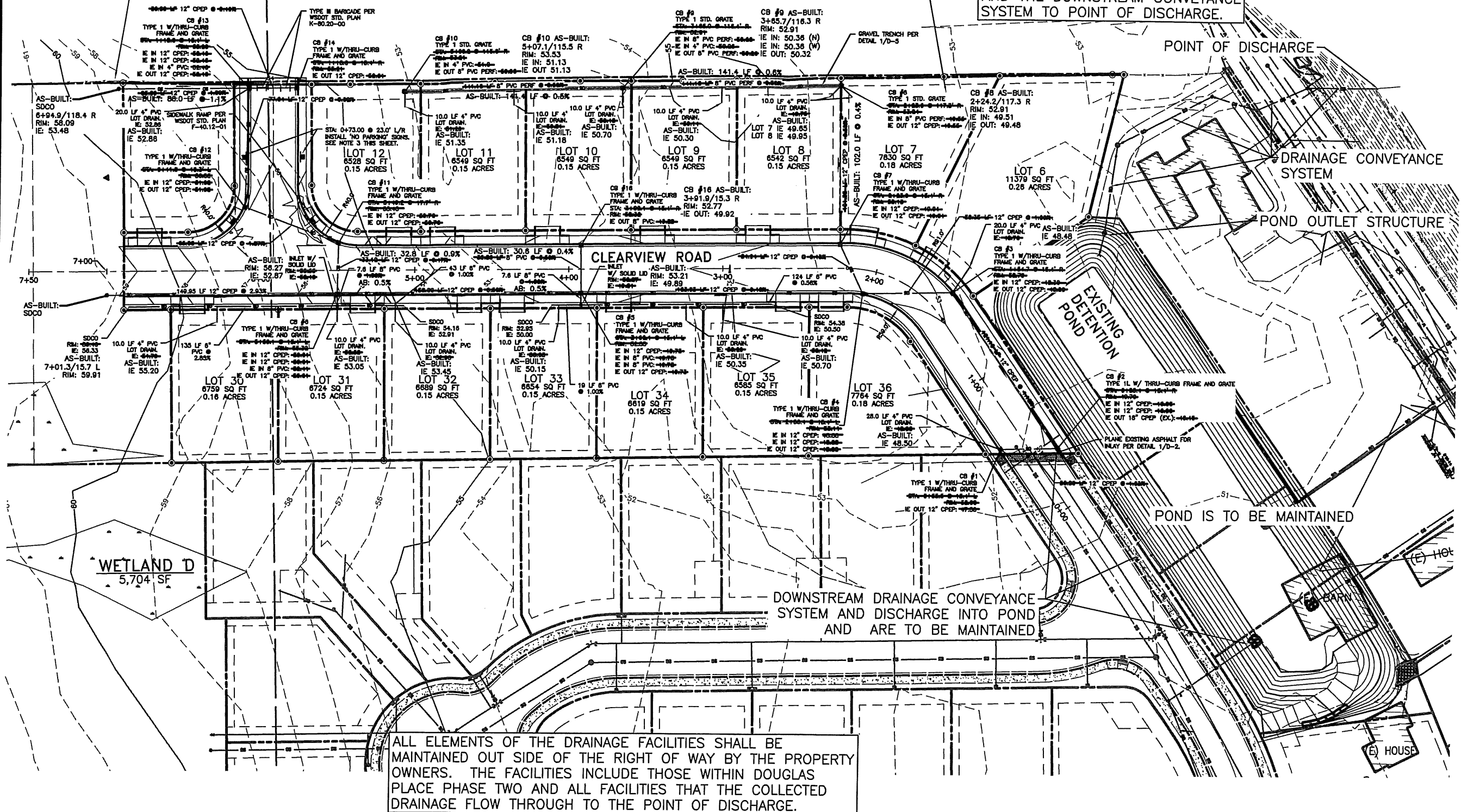
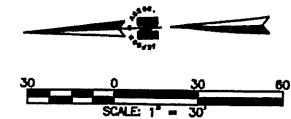
RONALD T. JEPSON & ASSOCIATES
REGISTERED PROFESSIONAL ENGINEERS AND LAND SURVEYORS
222 GRAND AVENUE, SUITE C, BELLINGHAM, WA 98225
PHONE: (360)733-5760 FAX: (360)647-8939
www.ronaldtjepson.com

PREPARED FOR: DOUGLAS PLACE LLC
GRAVEL TRENCH DETAIL
DOUGLAS PLACE PHASE 2
MAINTENANCE PLAN

SCALE: N.T.S.
DATE: 1/14
SHEET: **MN-03**
JOB #: 12082

THE DRAINAGE SYSTEM ALONG THE EAST SIDE OF LOTS AND THE DOWNSTREAM CONVEYANCE SYSTEM TO DISCHARGE LOCATION IS TO BE MAINTAINED IN ACCORDANCE WITH THE MAINTENANCE REPORT FOR THE DOUGLAS PALCE PHASE 2 REPORT

MAINTENANCE OF THE DRAINAGE SYSTEM INCLUDES THE POND, INLET STRUCTURES, OUTLET STRUCTURE AND THE DOWNSTREAM CONVEYANCE SYSTEM TO POINT OF DISCHARGE.



ALL ELEMENTS OF THE DRAINAGE FACILITIES SHALL BE MAINTAINED OUT SIDE OF THE RIGHT OF WAY BY THE PROPERTY OWNERS. THE FACILITIES INCLUDE THOSE WITHIN DOUGLAS PLACE PHASE TWO AND ALL FACILITIES THAT THE COLLECTED DRAINAGE FLOW THROUGH TO THE POINT OF DISCHARGE.

REVISIONS

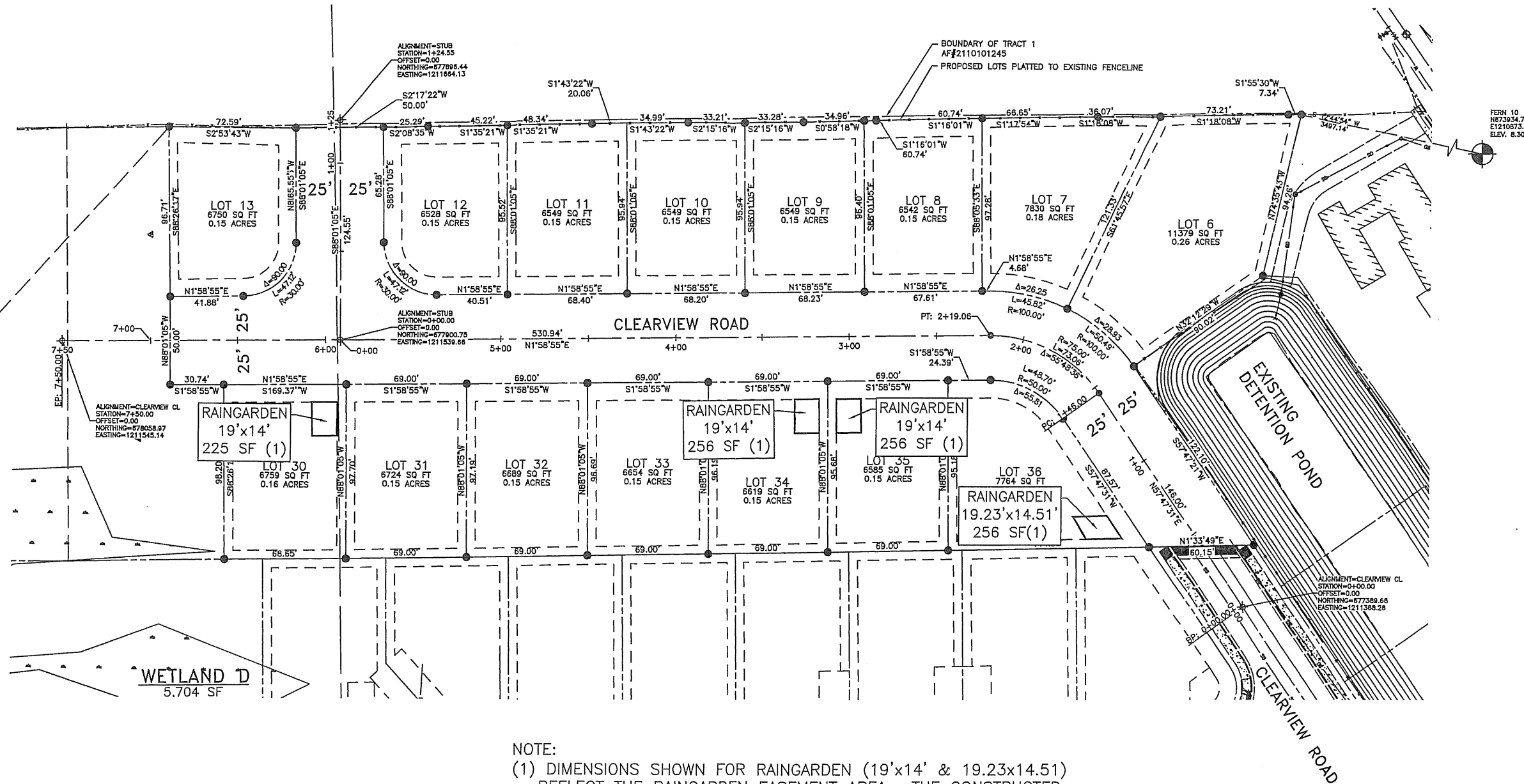
DESIGNED BY	RTJ
DRAWN BY	BJS
DATE	10/10/2013
ENGINEER	RONALD T. JEPSON
NO.	9381

PREPARED FROM AS-BUILT RECORD DRAWINGS

RONALD T. JEPSON & ASSOC.
 CIVIL ENGINEERING - SURVEYING - LAND PLANNING
 222 GRAND AVENUE, SUITE C, BELLINGHAM, WASHINGTON 98225
 360-733-5760 FAX 360-547-8839 WWW.JEPSONENGINEERING.COM

PREPARED FOR: DOUGLAS PLACE LLC

DRAINAGE SYSTEM MAINTENANCE SCHEMATIC
 DOUGLAS PLACE PHASE 2
 CLEARVIEW ROAD, FERNDALE WA
 COUNTY OF WHATCOMB



NOTE:
 (1) DIMENSIONS SHOWN FOR RAINGARDEN (19'x14' & 19.23x14.51)
 REFLECT THE RAINGARDEN EASEMENT AREA. THE CONSTRUCTED
 RAINGARDENS MUST MEET OR EXCEED THE SQUARE FOOTAGE AREAS
 SHOWN BELOW THE EASEMENT DIMENSIONS.
 LOT 30 - 225 S.F.
 LOT 34, 35 & 36 - 256 S.F.

REVISIONS	
DESIGNED BY	RTJ
DRAWN BY	BJS
DATE	1/30/2014
RONALD T. JEPSON ENGINEER	
REG.	9361

RECORD DRAWINGS

RONALD T. JEPSON & ASSOC.
 CIVIL ENGINEERING - SURVEYING - LAND PLANNING
 232 GRAND AVENUE, SUITE C, BELLINGHAM, WASHINGTON 98225
 360-735-5760 FAX 360-847-8839 WWW.JEPSONENGINEERING.COM

PREPARED FOR: DOUGLAS PLACE LLC
RAINGARDEN & SITE LAYOUT
 DOUGLAS PLACE PHASE 2
 CLEARVIEW ROAD, FERDALE WA
 COUNTY OF WHATCOM.
 WASHINGTON

SCALE:	NTS REDUCED
DATE:	1/14
SHEET:	MN-3
WD:	12082